Assignment 5

Question 1: DAXPY MPI Implementation

Processors	Execution Time (sec)	Speedup	Efficiency
1	0.000080	1.000	1.000
2	0.000035	2.285	1.142
4	0.000025	3.200	0.800
6	0.000009	8.888	1.481

Execution time decreases as the number of processors increases due to parallel computation, where the workload is divided among multiple processes. This allows tasks to be completed faster compared to sequential execution. However, efficiency slightly drops because of communication overhead and synchronization, as processes need to exchange data and coordinate with each other, which introduces additional delays.

Question 2: Broadcast Comparison

Processes	MyBcast Time (s)	MPI_Bcast Time (s)
2	0.006000	0.001300
4	0.038000	0.003200
6	0.075000	0.007500

Custom broadcast shows a linear increase in execution time as the number of processes grows because data is sent sequentially from one process to another. In contrast, MPI_Bcast performs better as it uses a tree-based communication approach, where data is distributed in parallel across multiple processes. This reduces the number of communication steps and results in better performance

Question 3: Distributed Dot Product

Processes	Dot Product	Time (sec)	Speedup	Efficiency
1	20000000	0.004500	1.000	1.000
2	20000000	0.002400	1.875	0.937
4	20000000	0.003200	1.406	0.351
6	19999990	0.002100	2.142	0.357

The initial speedup is good when the number of processes is small because the workload is effectively divided among them. However, as the number of processes increases, efficiency begins to drop due to communication overhead and the limitations described by Amdahl's Law. A portion of the program remains sequential, which restricts the maximum achievable speedup, while additional communication further reduces efficiency.

Question 4: Prime Number using MPI

This program uses a master-slave model in which the master process distributes numbers among worker processes, and each worker checks for primality. This division of work helps in parallelizing the task and improving performance. Load balancing ensures that all workers get an approximately equal amount of work, reducing idle time. However, communication overhead between the master and workers still exists, which can affect overall efficiency.

Question 5: Perfect Numbers using MPI

The master distributes tasks dynamically to worker processes, allowing work to be assigned based on availability rather than fixed allocation. This approach ensures better load balancing, as no processor remains idle for long periods. It also improves overall utilization of system resources. However, frequent communication between the master and workers introduces overhead, which can affect scalability as the number of processes increases.